

THE ATMOSPHERE OF SATURN: AN ANALYSIS OF THE VOYAGER RADIO OCCULTATION MEASUREMENTS

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ABSTRACT

During occultations of Voyager 1 and 2 by Saturn, the radio links from the spacecraft were utilized to probe the atmosphere of the planet at latitudes ranging from 36.7° N to 79.7° S. The measurements, which were conducted at wavelengths of 13 cm (*S* band) and 3.6 cm (*X* band), have yielded profiles in height of the electron number density in the ionosphere and the gas refractivity, number density, pressure, temperature, and ammonia abundance in the troposphere and stratosphere. The gas data were acquired in the pressure range from about 0.2 mbar to 1.3 bar over an altitude interval of 400 km. At the 1.3-bar level, the temperature was 145 ± 4 K and the lapse rate was equal to 0.8 K/km. The observed microwave absorption indicates that this region of the troposphere is saturated with ammonia. At the tropopause, which was detected between the 60- and 90-mbar levels, the temperature ranged from 80 K in the equatorial current to 87 K in the south polar region. The gas temperature above the tropopause increased with increasing altitude, reaching approximately 140 K near the 1-mbar level. In the equatorial current, the thermal structure of the stratosphere exhibits oscillations resembling vertically travelling planetary waves. From the vertical pressure profiles obtained at different latitudes, one may determine the size and shape of Saturn's isobaric surfaces. The 100-mbar surface has an equatorial radius of $60\,367 \pm 4$ km and a mean polar radius of $54\,438 \pm 10$ km. The corresponding oblateness $(R_{eq} - R_p)/R_{eq}$ is $0.098\,22 \pm 0.000\,18$. Relative to a reference geoid that has a polar radius of 54 438 km and whose shape is defined by the external gravity field and a uniform angular velocity corresponding to the system III rotation period, the observed 100-mbar isobaric surface has a 122 ± 11 km thick equatorial bulge produced by eastward zonal winds.

INTRODUCTION

Two Voyager spacecraft have explored the Saturn system (Stone and Miner 1981, 1982). The first, Voyager 1, was occulted by both Titan and Saturn, as well as the rings, during its flight through the system in November of 1980. The second spacecraft, which encountered Saturn in August of 1981, was also occulted by the planet. All occultations were observed with radio-tracking systems at NASA's Deep Space Network (Tyler *et al.* 1981, 1982).

The radio occultation experiments performed at Saturn utilized two coherently related tracking signals radiated from the spacecraft. Both signals, one at a frequency of 2.3 GHz (*S* band) and the other at 8.4 GHz (*X* band), were derived from a stable spacecraft oscillator. The transmission to the Earth was performed with a 3.66 m diam. parabolic dish rigidly fixed to the spacecraft. In order to keep the refracted beam pointed at the receiving station, the spacecraft was turned so that the antenna boresight tracked the virtual image of the station on Saturn's limb during the occultation measurements.

Tracking-station instrumentation included both wideband and phase-locked-loop receivers with digitized outputs. The main recording bandwidths used in the wideband receiver channels were 50 and 150 kHz at the *S* and *X* band, respectively. A 2-MHz backup recording system was also available. Because the wideband recordings provide the highest signal-to-noise ratios and allow detection of many signals appearing simultaneously at different frequencies during periods of multipath propagation, they are usually considered the primary data sets for atmospheric studies. The phase-locked-loop records, which contain samples of

the amplitude and frequency of the signal mode tracked by the receiver loop, are being used mostly for preliminary processing of ingress data when quick results are required during planetary encounter periods. More information on the radio instrumentation is available in earlier papers (Eshleman *et al.* 1977; Edelson *et al.* 1979).

The results from the early processing of the phase-locked-loop records were included in the preliminary team reports (Tyler *et al.* 1981, 1982). The main purpose of this paper is to summarize what we have learned from the wideband measurements of Saturn's atmosphere. In addition, we will briefly mention some new work on the data acquired at Titan.

DATA ANALYSIS AND RESULTS

The recording bandwidths employed in the wideband receiver channels were considerably greater than the frequency excursions produced by the atmosphere and the gravity field of Saturn. In order to reduce the bandwidths and the sample rates, the data were therefore filtered and decimated. Prior to detection, the signals were mixed with a variable local oscillator frequency which compensated for the changing Doppler shifts. The detection of the resulting audio band signals was accomplished with the help of computer software which simulates the operation of a phase-locked-loop receiver.

From the received signals, plus data on the spacecraft and station motion, one may determine the perturbations imposed on the tracking links by Saturn's atmosphere. Examples which show the effects of atmospheric absorption and refraction are provided in Figs. 1 and 2. These observations

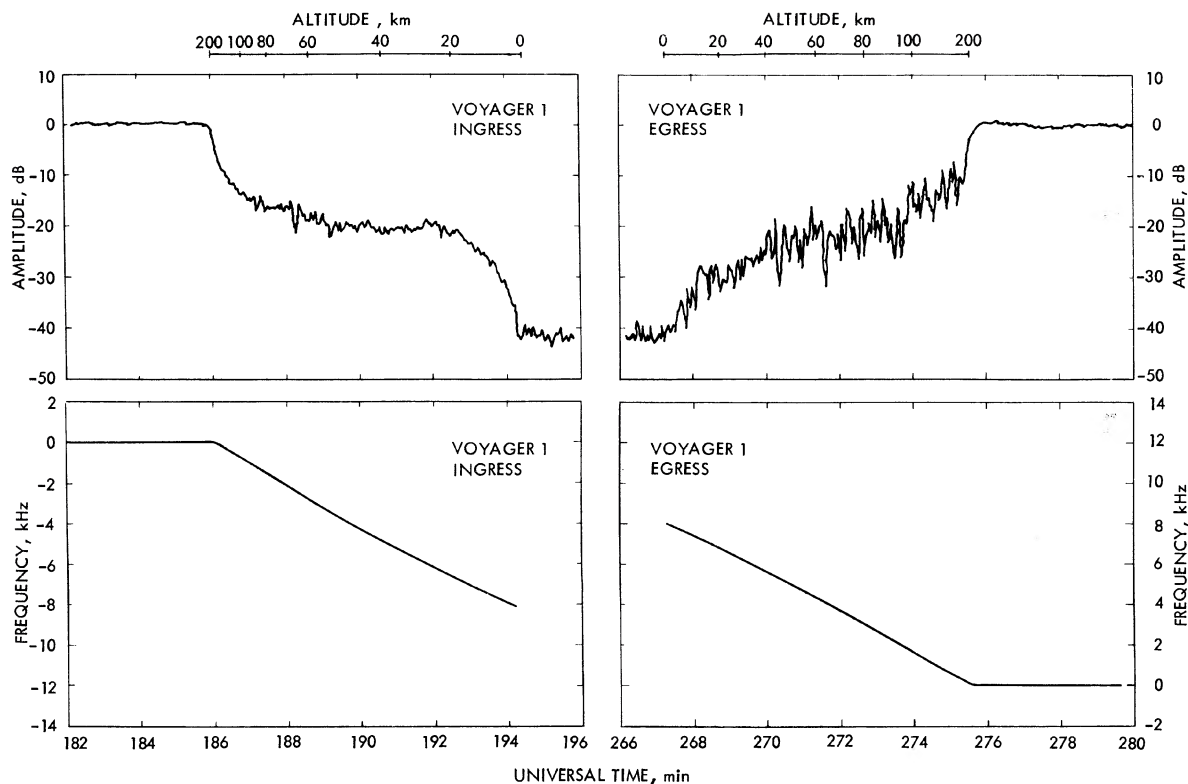


FIG. 1. Amplitude and Doppler frequency perturbations observed at X-band during the occultation of Voyager 1 by Saturn. Ingress occurred over Deep Space Station (DSS) 43 located in Australia, and egress over DSS 63 in Spain. The measurements were made on 13 November 1980. Time of signal reception is marked in minutes past midnight. Also indicated is the altitude of the link with respect to the 1-bar (or 10 N/cm²) pressure level. The drop in the signal intensity below the 200-km level was caused by defocusing and the second drop below 20 km was due to absorption. Turbulence or density waves in the equatorial jet caused strong amplitude scintillations during egress. The frequency rate was negative both during ingress and egress because the angle between the spacecraft's velocity vector and the ray to the Earth increased with time in both cases. Ingress took place in the south-polar region when the distance between the limb and the spacecraft was approximately 226 000 km. Egress occurred at a limb-to-spacecraft range of 304 800 km. The spacecraft was also occulted by Saturn's rings during egress (cf. Fig. 1 in Eshleman *et al.* 1977). However, in the time interval from 271 to 284 min past midnight, the link passed through the gap between the inner edge of the C ring and the planet. It took the signal about 1^h 24^m 47^s to travel from the spacecraft to the Earth.

may, in turn, be utilized to study the vertical structure of the atmosphere.

The next step in the analysis of the radio data consisted of inverting the integral equations relating the vertical absorption coefficient and refractive-index profiles to the signal perturbations observed during the occultations of the spacecraft. The mathematical solutions used in these computations are based on extensions of previous work (Fjeldbo and Eshleman 1968; Fjeldbo *et al.* 1971), and assume that the atmosphere is perfectly stratified with no horizontal variations within each of the four regions probed by the Voyager links. As indicated in the Appendix, the shape of the geodetic surfaces was obtained by integrating data on the external gravity field (Null *et al.* 1981) and the zonal winds observed at the cloud tops (Smith *et al.* 1982; Ingersoll and Pollard 1982).

Figure 3 shows the results of computing vertical refractivity profiles from the data acquired during Voyager 2's ingress and egress. The top scale in this figure gives the refractivity of the gas (N), which is defined as the refractive index less one times 10^6 . The ordinate scale indicates the local altitude of the measurements relative to the 1-bar level. When

the composition of the atmosphere is known, one may use the refractivity data to determine the number density distribution of the gas.

Infrared data acquired with the Pioneer 11 spacecraft indicated that Saturn's atmosphere consists of $90 \pm 3\%$ hydrogen and $10 \pm 3\%$ helium by number density, which corresponds to a mean molecular mass ($\bar{m}$) of 2.215 ± 0.060 amu (Orton and Ingersoll 1980). Subsequent measurements with Voyager yielded a hydrogen abundance of 94% with an uncertainty of a few percent—implying a depletion of helium in Saturn's atmosphere relative to that of Jupiter and the Sun (Hanel *et al.* 1981). A more recent interpretation of the spacecraft data has led to an atmospheric model containing 96.3% H₂, 3.3% He, and 0.4% CH₄ ($\bar{m} = 2.136$ amu) (Conrath *et al.* 1984). For the purpose of interpreting the refractivity data, we have adopted a hydrogen abundance of $94 \pm 3\%$; the remainder is assumed to be primarily helium. The corresponding mean molecular mass is 2.135 amu with an uncertainty of about 3%. For other gas mixtures, one may scale the results presented here.

By using the atmospheric composition indicated above and available experimental data relating refractivity to gas

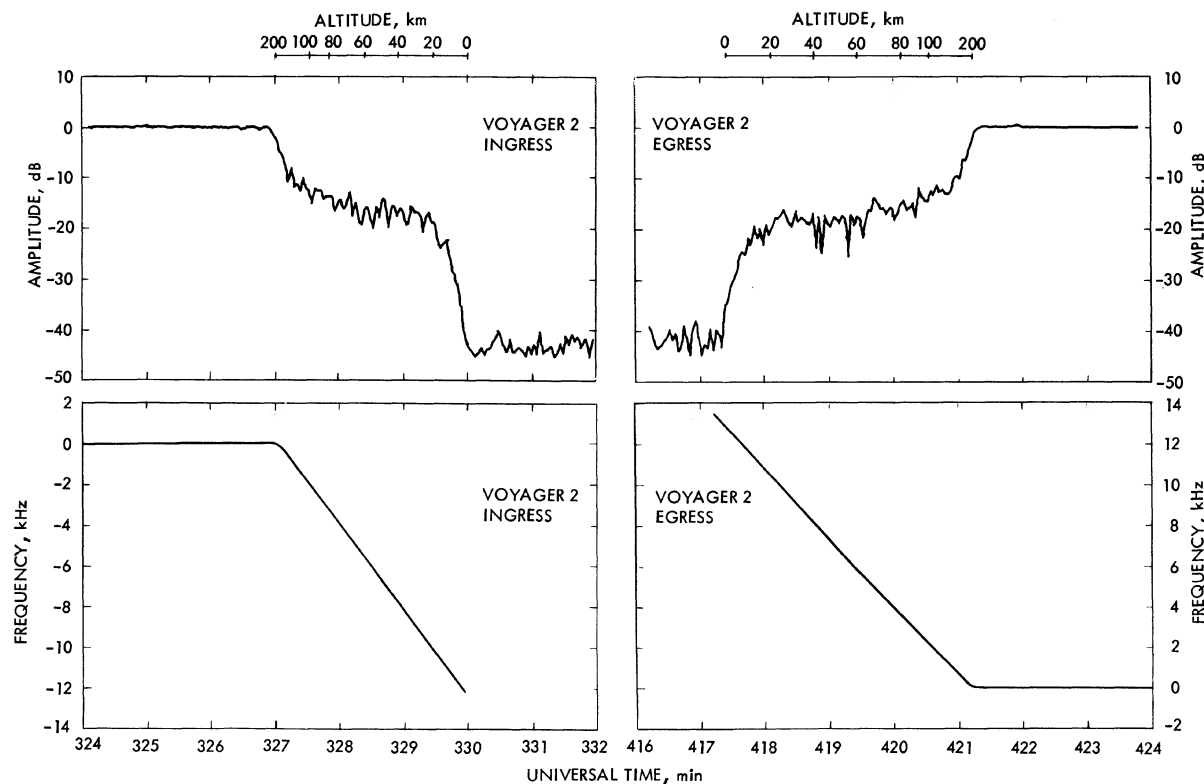


FIG. 2. Atmospheric perturbations of the X-band carrier signal received from Voyager 2 during the occultation of the spacecraft by Saturn. The data were acquired at DSS 43 on 26 August 1981. Time of signal reception is given in minutes past midnight. The propagation time was about $1^{\text{h}} 26^{\text{m}} 35^{\text{s}}$. Also shown is the altitude of the link relative to the 1-bar level. Ingress took place near 36° N planetographic latitude when the distance between Saturn's limb and the spacecraft was about 155 000 km. Egress occurred near 31° S latitude at a limb-to-spacecraft range of approximately 207 700 km. In order to obtain the highest possible signal-to-noise ratio during the occultation measurements, the spacecraft was maneuvered so that the antenna boresight was tracking the Earth's virtual image as viewed through the atmosphere of Saturn.

density (Essen 1953; Orcutt and Cole 1967; Bose *et al.* 1972), one may convert the refractivity measurements into number densities as illustrated by the bottom scale in Fig. 3. The resulting number density distributions were subsequently used to calculate the vertical pressure profiles by utilizing the equation for hydrostatic equilibrium and integrating the density from the top of the detectable atmosphere and downward. Finally, the equation of state was employed to compute the gas temperature from the number density and the pressure. The results are summarized in Figs. 4–6 and Table I.

Note that normal hydrogen, which consists of 75% *ortho* and 25% *para* H_2 (Farkas 1935), was used in the refractivity measurements conducted by Essen (1953) and by Orcutt and Cole (1967). When normal hydrogen from Saturn's deep atmosphere rises above the 3-bar level, the gas temperature drops below 200 K and some of the *ortho* H_2 may be converted to *para* H_2 . However, refractive-index data acquired at optical wavelengths indicate that the conversion of *ortho* to *para* does not produce a significant change in the refractive property of the hydrogen molecule. Thus, unknown variations in the *ortho*-to-*para* ratio should not have produced appreciable errors in the gas-state profiles derived from the radio occultation measurements.

The profiles computed from the radio data extend over a pressure range from about 0.2 mbar to 1.3 bar and an alti-

tude interval of 400 km. The height to which the gas could be detected in the stratosphere was limited by the stability of the spacecraft oscillators and by ionospheric scintillations which produced spectral broadening of the received signals. The depth of the measurements was limited by absorption. The X-band signals were extinguished below the 1-bar level and the S-band signals below 1.3 bar. As in earlier microwave studies of the giant planets (Gulkis and Poynter 1972; Berge and Gulkis 1976), we have assumed that the observed absorption was caused by ammonia vapor. On this basis, the absorption-coefficient measurements were utilized to infer the vertical distribution of NH_3 as illustrated in Fig. 3.

Voyager 2's radio signals probed Saturn's atmosphere at midlatitudes. The ingress gas data were acquired near the evening terminator in the region from 36.3° N planetographic latitude and 186.8° E system III longitude to 36.7° N latitude and 185.7° E longitude. The egress measurements were conducted in the early-morning atmosphere in the area from 31.2° S latitude and 69.9° E longitude to 31.5° S latitude and 71.7° E longitude. Figure 4 shows the temperatures computed from these measurements as a function of altitude. Also displayed is the standard deviation of the derived temperature due to the propagation of frequency noise, the uncertainty in the gas composition, plus wind velocity, gravity field, and ephemeris uncertainties. Systematic errors caused by horizontal refractivity variations cannot be determined

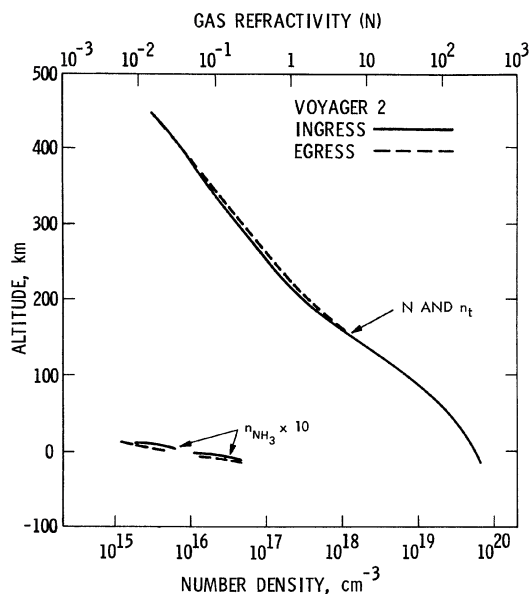


FIG. 3. Vertical gas refractivity and number density profiles deduced from radio recordings made during Voyager 2's occultation by Saturn. The gas refractivity (N), which is defined as the refractive index less 1 times 10^6 , was determined from Doppler frequency measurements at S and X bands. In computing the total gas density (n_t) from the refractivity, it was assumed that the atmosphere consists of 94% hydrogen with the remainder being mostly helium. For other nonpolar gas mixtures, one may scale the results presented here by noting that the number density is inversely proportional to the mean molecular refractivity. (The term "molecular refractivity" is here defined as the refractivity of the gas when the number density is equal to 1 cm^{-3} .) Also shown are the ammonia vapor density profiles (n_{NH_3}) inferred from the microwave absorption measurements. (The upper segment of each NH_3 profile was derived from X-band data and the lower segment from S-band data.) The ordinate scale gives the local altitude of the measurements relative to the 1-bar pressure level.

from the data. However, these errors are probably relatively small in the Voyager 2 profiles, since the spacecraft was setting and rising nearly perpendicular to Saturn's limb—thereby avoiding substantial latitudinal motion of the radio link during the atmospheric measurements.

Voyager 1 provided data in the south polar and the equatorial regions of Saturn. The ingress gas data were acquired in the late afternoon atmosphere along a 8600-km-long swath extending from 73.0° S , 43.1° E to 79.7° S , 27.9° E . Egress occurred near the morning terminator in the region from 1.5° S , 192.1° E to 8.9° S , 197.3° E . The vertical temperature profiles computed from these measurements are displayed in Figs. 5 and 6. While horizontal refractivity variations probably did not produce substantial systematic errors in the Voyager 2 profiles, the same cannot be said about the Voyager 1 results—since, in the latter case, the link traversed much larger horizontal distances. It should, furthermore, be noted that the Voyager 1 ingress profile, due to the high latitude of this measurement, is more sensitive to errors in the local zonal wind velocity than the other profiles. (This is because high-latitude velocity errors tilt the geodetic surfaces and thereby produce errors in the vertical refractivity profile when the spacecraft is setting obliquely on the limb. Near the 1-bar level in the south-polar profile, the sensitivity

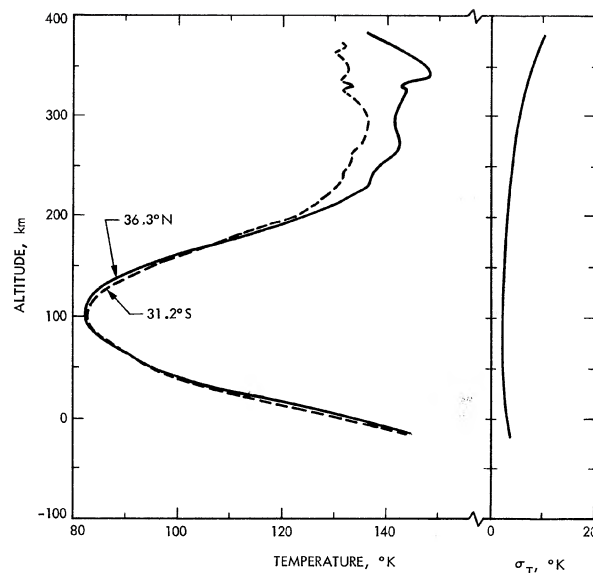


FIG. 4. Vertical temperature profiles computed from radio measurements conducted during Voyager 2's occultation by Saturn. Zero altitude corresponds to the 1-bar pressure level. Each profile has been labeled with the planetographic latitude at which the lowest point on the radio ray touched the 100-mbar isobaric surface. (The lowest point on the ray is defined as the position where the ray is perpendicular to the local vertical.) Also shown is the standard deviation of the computed temperature (σ_T). Oscillator frequency instabilities and spectral broadening due to ionospheric irregularities are the primary error sources in the stratosphere. Below the tropopause, the temperature errors reach a minimum and then increase again due to the reduction in the signal power and the effects of wind velocity uncertainties. The altitude resolution of the thermal profiles, as measured by the height of the first S-band Fresnel zone, ranged from about 10 km near the top of the stratosphere to 1 km at the 1.3 bar level. The corresponding resolution longitudinal to the link changed from about 1500 to 600 km over the same altitude interval.

of the gas temperature to errors in the eastward wind velocity is about -0.2 K s/m .) In addition, the Voyager 1 egress measurement was affected by strong atmospheric scintillations in the equatorial jet (cf. Fig. 1), attenuation in Saturn's C ring, and instrumentation problems at Deep Space Station 63, which resulted in loss of data records. We consequently regard the Voyager 1 profiles as less reliable than the Voyager 2 profiles.

The S-band links provided data down to the 1.3-bar pressure level, where the troposphere was saturated with ammonia vapor. The temperature and lapse rate at that level were 145 K and 0.8 K/km , respectively. As illustrated in Fig. 7, the maximum temperature lapse rate was observed near the 0.7-bar level. The drop in the lapse rate between 0.5 and 0.3 bar may mark a transition region between a lower convective haze layer having an adiabatic lapse rate and a higher stable regime in radiative equilibrium. In the region between 1.3 and 0.7 bar, the temperature lapse rate computed from the radio data is about 6% lower than the dry adiabatic rate expected from theory for a gas mixture consisting of 94% normal hydrogen and 6% helium. This difference may be due to the release of latent heat in rising gas when saturated NH_3 vapor is condensing into crystals and when *ortho* hydrogen is converted into *para* hydrogen through

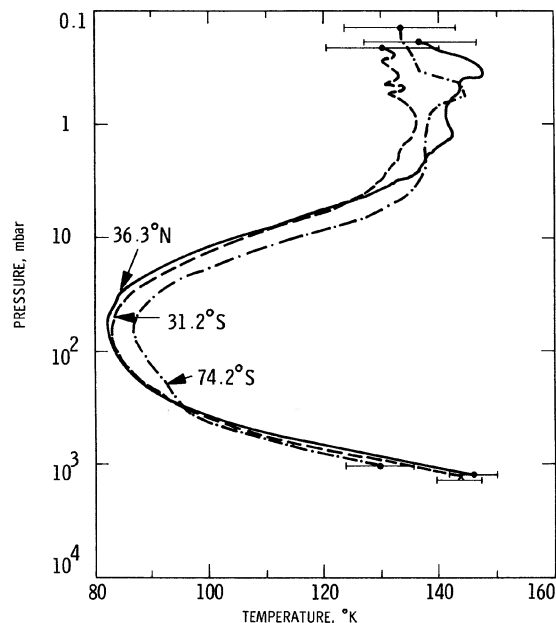


FIG. 5. Temperature profiles computed from radio refractivity data acquired at mid and high latitudes. The atmosphere was assumed to consist of 94% hydrogen and 6% helium by number density. For other gas mixtures, one may scale the results presented here. In a mixed atmosphere composed primarily of nonpolar molecules, one may perform the scaling by noting that the pressure is proportional to the mean molecular mass and inversely proportional to the mean molecular refractivity, and the temperature is approximately proportional to the mean molecular mass. The south-polar data show a warmer tropopause than the other measurements, which might be a seasonal effect. It was early autumn in the southern hemisphere of Saturn at the time of the Voyager encounters.

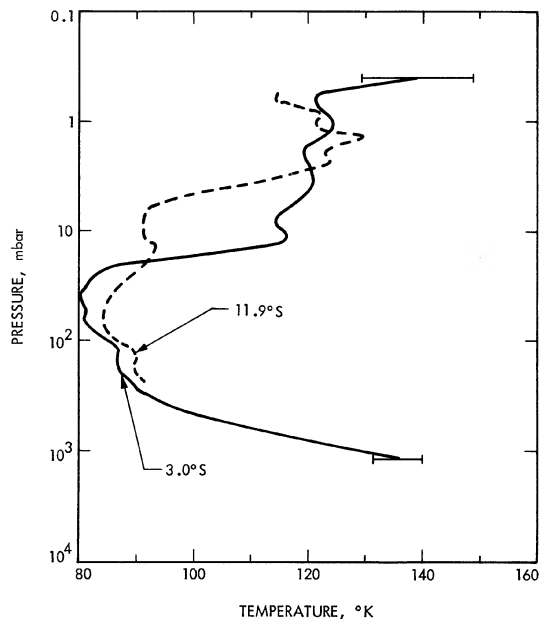


FIG. 6. Vertical temperature profiles for Saturn's equatorial current. The full drawn profile was obtained from measurements made during the egress of Voyager 1 near 3.0° S planetographic latitude. The stippled curve was derived from data acquired during Pioneer 11's egress near 11.9° S latitude (Kliore *et al.* 1980). Both measurements were made near the morning terminator. The structure of the profiles is suggestive of vertically propagating planetary waves. Similar effects have been observed in Jupiter's equatorial atmosphere (Lindal *et al.* 1981).

TABLE I. Atmospheric data.^a

p (mbar)	T (°K)	$n_{\text{NH}_3} \times 10^6/n_1$ (ppm)	h (km)	p (mbar)	T (°K)	$n_{\text{NH}_3} \times 10^6/n_1$ (ppm)	h (km)
0.20	138.7		376.7	173.78	87.9		71.6
0.25	143.2		363.9	199.53	89.8		66.9
0.32	147.3		350.7	229.09	91.4		62.0
0.40	146.2		337.4	251.19	92.4		58.7
0.50	143.8		324.3	275.42	93.5		55.4
0.63	142.2		311.4	302.00	94.8		52.0
0.79	141.2		298.6	331.13	96.5		48.6
1.00	141.6		285.8	363.08	98.4		45.1
1.26	142.5		273.0	398.11	100.7		41.6
1.58	140.3		260.2	436.52	103.2		37.9
2.00	137.7		247.7	478.63	106.0		34.2
2.51	136.9		235.3	501.19	107.5		32.3
3.16	133.4		223.1	524.81	109.1		30.3
3.98	129.0		211.3	549.54	110.7		28.4
5.01	123.5		199.9	575.44	112.4		26.4
6.31	117.7		189.0	602.56	114.1		24.4
7.94	112.0		178.7	630.96	115.8		22.3
10.00	104.8		168.9	660.69	117.6		20.2
12.59	99.5		159.8	691.83	119.4		18.1
15.85	95.0		151.0	724.44	121.3		16.0
19.95	90.5		142.8	758.58	123.2		13.8
23.99	87.6		136.4	794.33	125.0		11.6
28.84	85.3		130.2	831.76	127.0	2.6	9.3
34.67	83.9		124.1	870.96	128.9	5.1	7.0
41.69	83.1		118.1	912.01	130.8	7.1	4.7
50.12	82.3		112.2	954.99	132.8	9.5	2.4
60.26	82.0		106.3	1000.00	134.8	10.9	0.0
72.44	82.4		100.4	1047.13	136.8		-2.4
87.10	82.7		94.4	1096.48	138.9	21.2	-4.9
100.00	83.4		90.0	1148.15	140.9	41.9	-7.4
114.82	84.3		85.5	1202.26	142.9	49.2	-9.9
131.83	85.2		80.9	1258.93	145.0	66.9	-12.5
151.36	86.3		76.3	1298.48	146.2		-14.1

^a This table gives the temperature (T), and the ammonia mixing ratio ($n_{\text{NH}_3} \times 10^6/n_1$) in parts per million as a function of the pressure level (p) in Saturn's atmosphere. Also provided is the local altitude (h) of the measurements relative to the 1-bar pressure level. The gas is assumed to consist of 94% H_2 with the remainder being He and NH_3 . For other gas mixtures, one may scale the numerical data given here. The measurements were made during the ingress of Voyager 2.

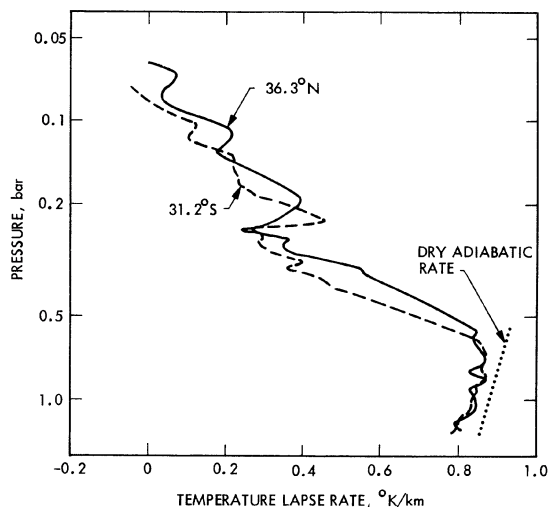


FIG. 7. Temperature lapse rates computed from the radio data as a function of pressure level in the troposphere of Saturn. Also shown is the dry adiabatic lapse rate expected from theory for normal hydrogen, i.e., when the *ortho*-to-*para* ratio is equal to 3. As discussed in the text, convection may be the dominant mechanism of vertical heat transport below 0.5 bar and radiative transfer may dominate above the 0.3-bar level.

catalytic interaction with the NH_3 crystals. Calculations performed by Massie and Hunten (1982) show that the *ortho*-to-*para* conversion alone may reduce the adiabatic temperature lapse rate in Saturn's aerosol layers by up to 10%.

As illustrated in Fig. 5 and 6, the temperature profiles show relatively little change with latitude in the 0.6–1.3-bar pressure range. Above the 0.6-bar level, the south-polar data show higher tropospheric temperatures than the other data sets. This may be a seasonal effect. Similar effects were also noticed in the infrared data acquired with Voyager (Conrath *et al.* 1984). At the tropopause, which was observed between the 60- and 90-mbar levels, the temperature ranged from 80 K in the equatorial current to 87 K in the south-polar area.

In the region above the tropopause, the temperature was increasing with altitude reaching approximately 140 K near the 1-mbar level in the stratosphere. The thermal profiles computed for the equatorial jet (cf. Fig. 6) exhibit vertical oscillations similar to those derived for Jupiter's equatorial stratosphere—which have variously been interpreted in terms of vertically propagating Rossby waves, nonuniform aerosol heating, and high- and low-pressure regions produced by local atmospheric dynamics (Allison 1983; Appleby and Hogan 1984; Lindal *et al.* 1981). Above the 0.2-mbar level, the gas temperature may decrease again. A study of the ultraviolet data obtained during the Saturn encounter indicated a 120 ± 30 K temperature at a pressure of 0.05 μbar (Smith *et al.* 1983).

At ionospheric heights, the radio data may be used to study the vertical distribution of free electrons. Figure 8 shows the electron number density profiles obtained by inverting the dispersive Doppler data acquired with Voyager. Due to systematic errors caused by horizontal plasma variations and multipath propagation in the lower layers, the accuracy of the profiles decreases with decreasing altitude. As in earlier studies of Jupiter, the results show large differences between the regions sampled by the links—indicating that local processes and events play an important role in forming

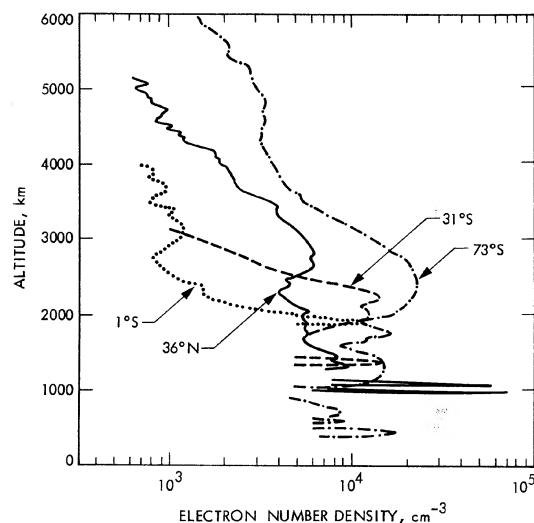


FIG. 8. Vertical electron number density profiles for Saturn's ionosphere. The ordinate scale gives the local altitude relative to the 1-bar level. The data show many sharp ionization layers which may have been produced by vertical wind shear (Chen 1981).

the layers (Ashihara and Shimizu 1977; Chen 1981; Connerney and Waite 1984).

Finally, we have utilized the vertical pressure profiles to study the size and shape of Saturn (Lindal *et al.* 1984). The results, which provide useful boundary conditions in modeling the interior of the planets (Hubbard and Smoluchowski 1973; Busse 1975; Stevenson and Salpeter 1976; Zharkov and Trubitsyn 1976; Horedt and Hubbard 1983; Hubbard and Horedt 1983), are summarized in Fig. 9. The five data points shown in the left-hand portion of this figure give the observed radius of the 100-mbar isobaric surface as a function of planetocentric latitude. Four of the five points shown are from the Voyager occultation measurements, and the last point was obtained from an analysis of radio-tracking data acquired with Pioneer 11. The solid curve represents the level surface or geoid of a rotating fluid which best fits the radio data. As outlined in the Appendix, the geodetic surface was obtained by integrating equation (8) using data on the external gravity field and the zonal wind velocities observed at the cloud tops. In order to perform the integration, it was assumed that the wind velocities are independent of altitude. This assumption seems to be justified by the results of the calculations, however, since all the isobaric surfaces in the troposphere appear to fit the computed geoids about equally well—the standard deviations of these fits range from 3 to 4 km—although Conrath and Pirraglia (1983) have found that there may be significant vertical thermal wind shear near the 150-mbar level at some latitudes.

The isobaric and geodetic surfaces agree within about 4 km in the troposphere. Thus, by comparing the radio data with the computed geoids, one may estimate the radius of an isobaric surface at any latitude. For instance, the geodetic surface which in a least-squares sense best fits the 100-mbar data has a mean polar radius of $54\,438 \pm 10$ km, and an equatorial radius of $60\,367 \pm 4$ km. The oblateness of this surface $(R_{eq} - R_p)/R_{eq}$ is $0.098\,22 \pm 0.000\,18$. The corresponding values at the 1-bar level are $54\,364 \pm 10$ km, $60\,268 \pm 4$ km, and $0.097\,96 \pm 0.000\,18$.

The right-hand portion of Fig. 9 illustrates the effect that

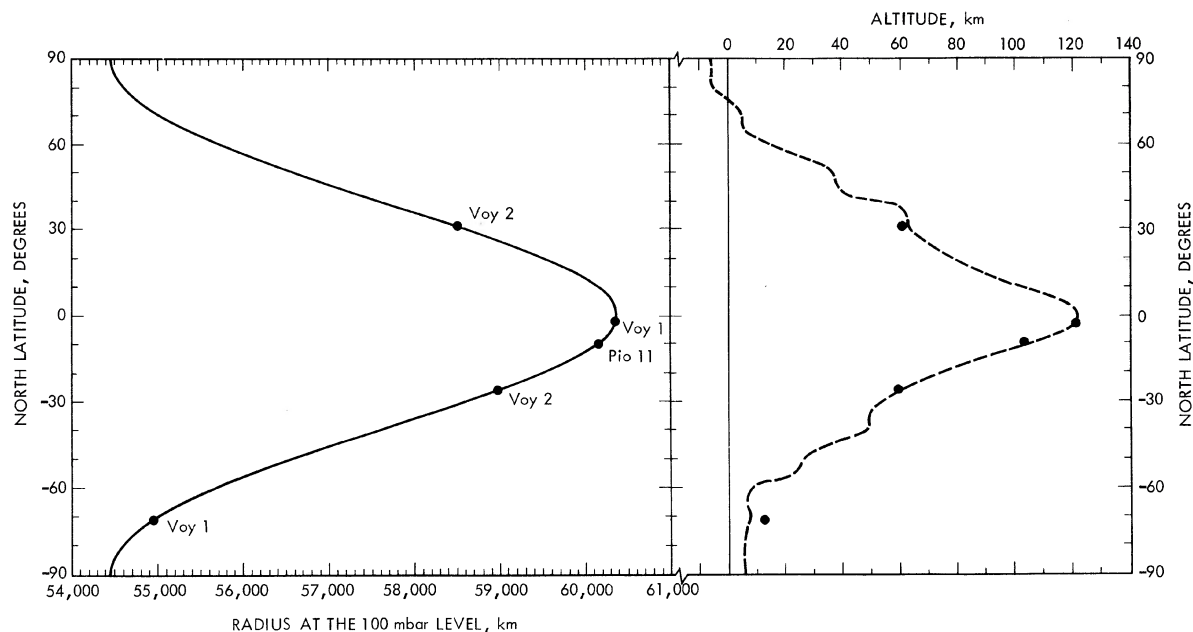


FIG. 9. Saturn's physical shape. The discrete points in the left-hand portion of the chart show the radius of the 100-mbar isobaric surface as a function of planetocentric latitude. The full drawn curve represents the horizontal surface or geoid which best fits the radio data. At the 100-mbar level in Saturn's atmosphere, the equatorial radius is $60\,367 \pm 4$ km and the mean polar radius is $54\,438 \pm 10$ km. (The south-polar radius may be of the order of 10 km greater than the north-polar radius—possibly a seasonal atmospheric effect.) The right-hand portion of the figure shows the observed altitude of the 100-mbar isobaric surface (discrete points) and the corresponding best-fit geoid (stippled curve) computed from zonal-wind data acquired with the Voyager imaging system (Smith *et al.* 1982; Ingersoll and Pollard 1982; Garneau 1984). Altitudes are measured relative to a reference geoid that has a polar radius of 54 438 km and whose shape is defined by the external gravity field and a uniform angular velocity given by the $10^{\text{h}}\,39^{\text{m}}\,22.4^{\text{s}}$ magnetic field or system III rotation period (Desch and Kaiser 1981; Davies *et al.* 1983).

the zonal winds have on Saturn's physical shape. The discrete points give the height of the 100-mbar level with respect to a reference geoid that has a polar radius of 54 438 km and whose shape is defined by the external gravity field and a uniform angular velocity corresponding to the magnetospheric rotation period. Also shown is the geoid (obtained by utilizing zonal wind data) which best fits the radio measurements. Relative to the reference geoid, the observed 100-mbar surface has an equatorial bulge of 122 ± 11 km.

DISCUSSION

An earlier study of Jupiter did not show a detectable zonal-wind bulge in the isobaric surface data (Lindal *et al.* 1981). The reason for the greater equatorial bulge on Saturn may be understood by studying equation (18) in the appendix and by noting that the acceleration of gravity is a factor of 2.6 smaller and the eastward zonal-wind velocities are about five times greater on Saturn than on Jupiter. Also contributing to the difference is the fact that Saturn's equatorial jet extends to higher latitudes and that the zonal winds reverse their direction to a lesser extent than is the case on Jupiter.

Dynamical considerations suggest that the zonal winds observed at Saturn's cloud tops do not abruptly die out a couple of scale heights below the 1.3-bar level—because that would imply that the 10-bar geoid would approach the 1.3-bar geoid in the polar regions and thereby produce very large vertical pressure gradients at these latitudes. It, therefore, seems reasonable to assume that the winds continue to blow to a depth of many atmospheric scale heights before dying out. Based on a study of imaging and infrared data acquired

with Voyager, Smith *et al.* (1982) have suggested that the winds blow down to at least 10 000 bar.

The infrared instrument team has studied the latitudinal variations of the gas temperature at three different pressure levels in Saturn's troposphere (Conrath *et al.* 1984). A comparison between the radio and infrared results is provided in Fig. 10. As may be seen from this figure, the temperatures derived from the two experiments agree within 3 K—suggesting that the magnitude of the total error (systematic errors plus noise) does not exceed this value in either experiment for the 110–730-mbar pressure range.

The work on the ammonia distribution in Saturn's troposphere has, together with revised data on the spacecraft turns performed during the encounter period, led to an improved model for the X-band radiation field of the high-gain spacecraft antenna. These changes also affect the interpretation of the microwave-loss data acquired at Titan (Lindal *et al.* 1983). In particular, it has resulted in a revised estimate for the absorption observed when the X-band link first cleared Titan's surface during egress. The difficulty with that absorption measurement is that the spacecraft high-gain antenna was pointed about 0.3° away from the virtual image of the Earth as viewed through Titan's atmosphere. Thus, relatively small errors in the antenna radiation pattern and the spacecraft orientation can produce significant errors in the effective antenna gain and, therefore, the total absorption computed for the ray grazing the surface of the satellite. The earlier analysis yielded an upper limit to the X-band absorption encountered through Titan's lower troposphere of about 2 dB. The new work indicates a loss of 3 ± 3 dB, which incidentally is in better agreement with absorption

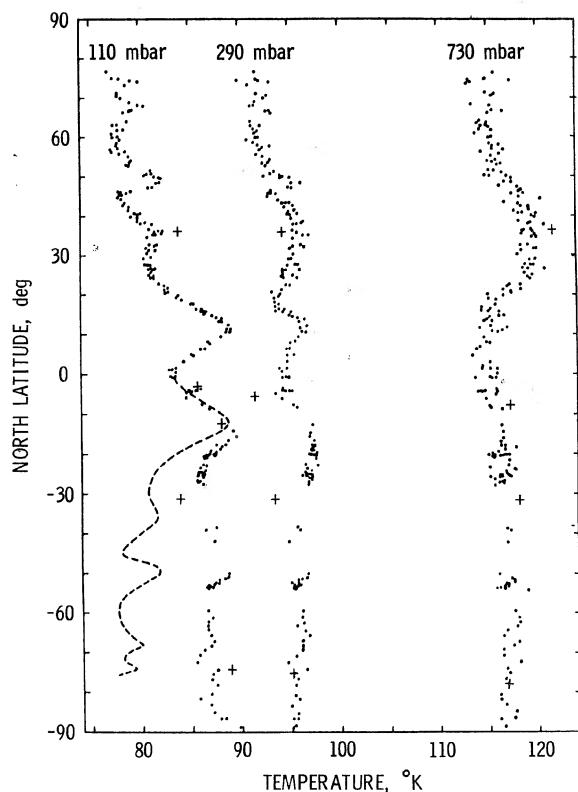


FIG. 10. Atmospheric temperatures at the 110-, 290-, and 730-mbar pressure levels as a function of planetographic latitude. The dots represent data acquired with the infrared instruments (IRIS) carried on board the Voyager spacecraft (Conrath *et al.* 1984), and the crosses show the radio data discussed in this report. The stippled curve gives the infrared temperatures observed at the 110-mbar level in the northern hemisphere, but plotted at southern latitudes.

expected from nitrogen when the results of Ho *et al.* (1966) are extrapolated to the 94-K surface temperature of Titan (Muhleman *et al.* 1984). The refractivity and S-band loss profiles were not affected by the new work, however.

TABLE II. Physical data on Jupiter and Saturn.

	Jupiter	Saturn
System III rotation period (Davies <i>et al.</i> 1983)	9hr 55min 29.7s	10hr 39min 22.4s
At the 1-bar pressure level:		
Radius:		
Equatorial	71 492 $\pm$ 4 km	60 268 $\pm$ 4 km
Polar (mean)	66 854 $\pm$ 10 km	54 364 $\pm$ 10 km
Oblateness:		
$(R_{eq} - R_p)/R_{eq}$	0.064 87 $\pm$ 0.000 15	0.097 96 $\pm$ 0.000 18
Acceleration of gravity:		
Equatorial ^a	23.12 $\pm$ 0.01 m/s ²	8.96 $\pm$ 0.01 m/s ²
Polar	27.01 $\pm$ 0.01 m/s ²	12.14 $\pm$ 0.01 m/s ²
Atmospheric temperature ^b	165 $\pm$ 5 K	134 $\pm$ 4 K

^a Values apply to objects corotating with the atmospheric gas.

^b Values assume that the atmospheres of Jupiter and Saturn consist of 89% and 94% hydrogen, respectively—with the remainder being helium and ammonia.

Based on a study of the temperature lapse rates computed from the radio occultation data acquired at Titan, Lindal *et al.* (1983) have derived an upper limit of about 3% for the methane composition in the lower troposphere of the satellite. This result was obtained by assuming that the lower troposphere is perfectly mixed below the CH₄ condensation level and by determining the minimum methane cloud altitude that does not produce radio lapse rates greater than the wet adiabatic rate in the saturated region. The wet adiabatic-temperature rate was calculated by using a formula published by Chamberlain (1978). However, formulas published by Lasker (1963) and Houghton (1977) give a somewhat higher value for the wet adiabatic lapse rate than Chamberlain's expression (Eshleman *et al.* 1983; Lunine *et al.* 1983). When the study of the radio temperature lapse rates was repeated using Houghton's formula, the minimum methane saturation level was reduced from 17 to about 14 km; however, the upper limit for the CH₄ content below the condensation level remained approximately 3%.

A summary of the more significant results obtained from the Saturn data set is given in Table II. For comparison, we have also included earlier results from Jupiter.

The radio data discussed in this article were acquired through the coordinated efforts of the Voyager Project staff, the Deep Space Network personnel, the Radio Science Support Team, and the Radio Science Team. The support team consisted of B. J. Buckles, P. E. Doms, D. F. Finnerty, D. P. Holmes (team leader during Voyager 2's encounter with Saturn), D. N. Sweetnam, and G. E. Wood (team leader during Voyager 1's encounter with Saturn), all from the Jet Propulsion Laboratory. The scientists appointed by the National Aeronautics and Space Administration to conduct radio propagation experiments with the Voyager spacecraft consist of J. D. Anderson, G. S. Levy, G. F. Lindal, and G. E. Wood from the Jet Propulsion Laboratory, T. A. Croft from SRI International, and V. R. Eshleman and G. L. Tyler (team leader during the Saturn encounters) from Stanford University. We thank the Voyager Navigation Team for providing the ephemeris data needed to analyze the occultation measurements and H. B. Hotz, E. R. Kursinski, and P. H. Richter for help with data processing related tasks. The research described in this paper was carried out at the Jet Propulsion Laboratory of the California Institute of Technology and at Stanford University under contracts with the National Aeronautics and Space Administration.

APPENDIX: GEODETIC CALCULATIONS

The purpose of this Appendix is to briefly outline how the shape of Saturn's geoid may be determined from data on the gravitational field and the zonal-wind velocities.

Tracking data acquired with the Pioneer and Voyager spacecraft indicate that Saturn's external gravity field is symmetric with respect to the spin axis, as one would expect for a fluid in hydrostatic equilibrium (Null *et al.* 1981; Campbell 1984). Thus, the acceleration produced by the external gravity field can be regarded as a vector contained in the local meridian plane.

For an object corotating with the atmospheric gas, one must include the effects of both the external gravity field and the centrifugal force when computing the acceleration of gravity. The resulting acceleration (g) may be expressed in terms of its radial (g_r) and latitudinal (g_ϕ) components

$$g = u_r g_r + u_\phi g_\phi, \quad (1)$$

where $\mathbf{u}_r$ and $\mathbf{u}_\phi$ are the unit vectors in the radial and latitudinal directions, respectively. The vector components are given by

$$g_r(r, \phi) = -\frac{GM}{r^2} + \frac{GM}{r^2} \sum_{i=1}^{\infty} (2i+1) J_{2i} \left(\frac{R}{r}\right)^{2i} P_{2i}(\sin \phi) + \frac{2}{3} \omega^2 r [1 - P_2(\sin \phi)], \quad (2)$$

$$g_\phi(r, \phi) = -\frac{GM}{r^2} \sum_{i=1}^{\infty} J_{2i} \left(\frac{R}{r}\right)^{2i} \frac{dP_{2i}(\sin \phi)}{d\phi} - \frac{1}{3} \omega^2 r \frac{dP_2(\sin \phi)}{d\phi}, \quad (3)$$

where G is the universal gravitational constant, M is the mass of Saturn, J_{2i} is the $2i$ th zonal harmonic coefficient, R is the radius to which the zonal harmonic coefficients have been normalized, P_{2i} is the Legendre polynomial of degree $2i$, and ω is the angular velocity. The independent variables r and ϕ denote the distance to the center of mass and the planetocentric latitude, respectively.

For ω , we have

$$\omega = \omega_{\text{III}} + \frac{V_w}{r \cos \phi}, \quad (4)$$

where V_w is the eastward zonal-wind velocity and ω_{III} is the angular velocity corresponding to the system III rotation period. Published zonal-wind data (Smith *et al.* 1982; Ingersoll and Pollard 1982) give V_w as a function of the planetographic latitude, which simply is equal to $(\phi + \psi)$, where

$$\psi = \arctan(g_\phi/g_r). \quad (5)$$

Of course, V_w may also change with altitude. However, the altitude variation was assumed to be negligible. This simplification seems to be justified by the results of the analysis—at least in the troposphere, where the computed geoids appear to fit the isobaric surfaces about equally well at all observed pressure levels.

Equations (1)–(5) together with data on the external gravity field (Null *et al.* 1981; Campbell 1984), the zonal-wind velocities (Smith *et al.* 1982; Ingersoll and Pollard 1982), and the pole vector (Kozai 1957; Null *et al.* 1981; Simpson *et al.* 1983) allow the acceleration of gravity ($\mathbf{g}$) and, therefore, the local vertical to be determined at any point along a radio ray in Saturn's atmosphere.

The differential equation for a local vertical or field line is

$$d\phi = \frac{1}{r} \tan \psi \, dr, \quad (6)$$

which may be integrated from any point of interest with the help of a digital computer. For an elemental change in the altitude (dh) measured along a field line, one has

$$dh = dr / \cos \psi. \quad (7)$$

Thus, the altitude difference between two points on the same field line is obtained by simultaneously integrating equations (6) and (7).

A geoid may be defined as a surface that is everywhere orthogonal to the local vertical. Because of symmetry about the spin axis, it is sufficient to consider the radius (r) of a geoid a function only of the planetocentric latitude (ϕ). The differential equation for a geoid is

$$dr = -\tan \psi \, r \, d\phi, \quad (8)$$

which may be integrated to yield the shape ($r(\phi)$) of any geoid of interest.

For the purpose of studying the equatorial bulge produced by the eastward zonal winds (cf. Fig. 9), it is also useful to define a reference geoid in terms of no zonal winds. The shape of the reference geoid ($r_{\text{ref}}(\phi)$) may, of course, be obtained by substituting ω_{III} for ω and integrating equation (8). However, since in this case we are dealing with a curl free field, it is, as outlined below, somewhat more convenient to perform the calculations with the help of the scalar-potential field theory.

For a uniformly rotating fluid, the gravity potential ($U(r, \phi)$) may be defined in terms of

$$\mathbf{g} = -\text{grad}(U). \quad (9)$$

The potential may be obtained by integrating $\mathbf{g}$ from the point of interest (r, ϕ) to a point at infinity on the spin axis:

$$U(r, \phi) - U(\infty, \pi/2) = \int_{(r, \phi)}^{(\infty, \pi/2)} \mathbf{g} \, d\mathbf{s}, \quad (10)$$

where the infinitesimal vector $d\mathbf{s}$ represents an elemental displacement along the path of integration. For a fluid with a uniform angular velocity corresponding to the system III rotation period (ω_{III}), one obtains the well-known formula

$$U(r, \phi) = -\frac{GM}{r} + \frac{GM}{r} \sum_{i=1}^{\infty} J_{2i} \left(\frac{R}{r}\right)^{2i} P_{2i}(\sin \phi) - \frac{1}{2} \omega_{\text{III}}^2 r^2 \cos^2 \phi, \quad (11)$$

where the constant of integration has been chosen so that $U(r, \phi)$ is equal to zero on the spin axis at infinity.

In what follows, we will outline how equation (11) may be used in an iterative manner to calculate the shape of a reference geoid ($r_{\text{ref}}(\phi)$) with polar radius r_p . For the purpose of illustrating the method, we will describe the computation of the 100-mbar reference geoid utilized in Fig. 9. The first step in this calculation consisted of determining the gravity potential (U_{ref}) of the reference geoid from equation (11):

$$U_{\text{ref}} = U(r_p, \pi/2), \quad (12)$$

where r_p was set equal to 54 438 km. Next, an initial value was assigned to $r_{\text{ref}}(\phi)$ and the corresponding potential $U(r_{\text{ref}}, \phi)$ was determined from equation (11). The error in this potential (ΔU) was then calculated:

$$\Delta U = U(r_{\text{ref}}, \phi) - U_{\text{ref}}. \quad (13)$$

ΔU was subsequently utilized to compute the correction required in the radius (Δr) at the latitude ϕ :

$$\Delta r = \Delta U / g_r, \quad (14)$$

where g_r was obtained by substituting ω_{III} for ω in equation (2). The procedure was then repeated with the new value for the radius ($r_{\text{ref}}(\phi) + \Delta r$). These computations were continued until the desired accuracy was achieved for $r_{\text{ref}}(\phi)$.

Figure 11 illustrates how eastward zonal winds perturb the geoid. The altitude of a geoid ($h(\phi)$) relative to the reference geoid with the same polar radius (r_p) is obtained by integrating equations (6) and (7) as indicated above. In what follows, we shall look at some approximations that may be useful when studying $h(\phi)$.

A zonal jet with an eastward velocity V_w perturbs the acceleration of gravity at a point (r, ϕ) by an amount ($|\mathbf{g} - \mathbf{g}_{\text{ref}}|$) given by

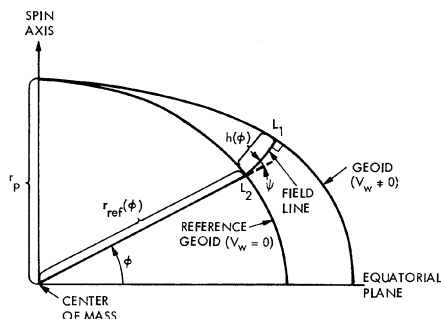


FIG. 11. Geometric properties of geodetic surfaces.

$$\Delta g_w = V_w \left[2\omega_{III} + \frac{V_w}{r \cos \phi} \right], \quad (15)$$

where Δg_w may be thought of as a vector in the meridian plane parallel to the equatorial plane. By considering the line integral of g from a location L_1 on the geoid (cf. Fig. 11) down the local vertical (or field line) to point L_2 on the reference geoid and from there along the reference geoid to the spin axis, one may show that

$$\int_{\phi}^{\pi/2} \Delta g_w \sin(\phi + \psi_{\text{ref}}) \frac{r_{\text{ref}}(\phi) d\phi}{\cos \psi_{\text{ref}}} = \int_0^{h(\phi)} |g| dh, \quad (16)$$

where ψ_{ref} may be computed from equations (2), (3), and (5) by substituting ω_{III} for ω . (Note that equation (16) is strictly correct only when deviations from hydrostatic equilibrium and horizontal stratification are negligible at all geodetic levels traversed by the path of integration). Finally, one may solve equation (16) with respect to the altitude of the geoid

$$h(\phi) = \frac{1}{\langle g \rangle} \int_{\phi}^{\pi/2} V_w \left[2\omega_{III} + \frac{V_w}{r_{\text{ref}}(\phi) \cos \phi} \right] \frac{\sin(\phi + \psi_{\text{ref}})}{\cos \psi_{\text{ref}}} r_{\text{ref}}(\phi) d\phi, \quad (17)$$

which, for small wind velocities and a negligible oblateness, reduces to the well-known result

$$h(\phi) \approx \frac{2\omega_{III} r_p}{\langle g \rangle} \int_{\phi}^{\pi/2} V_w \sin \phi d\phi. \quad (18)$$

Here, $\langle g \rangle$ denotes the average acceleration of gravity along the field line connecting points L_1 and L_2 . Note that $\langle g \rangle$ is a slowly varying function of $h(\phi)$. Thus, equations (17) and (18) are well suited for determining $h(\phi)$ in an iterative manner on a computer.

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